Machine Learning for Mechanical Engineers at CoEP Session 16: PCA

Session 16: PCA

PCA

Too much of anything is good for nothing!

What happens when a data set has too many variables ? Here are few possible situations which you might come across:

- You find that most of the variables are correlated.
- You lose patience and decide to run a model on whole data. This returns poor accuracy and you feel terrible.
- You become indecisive about what to do
- You start thinking of some strategic method to find few important variables

More number of variables means High Dimensionality.

Curse of Dimensionality

- Data-sets typically high dimensional
- Images of 20x20 bitmaps have 400 dimensions. Mega pixel images has far too much.
- In text, all words are features, say, 10^6.

Curse of Dimensionality

- Machine Learning methods are statistical, and data sparse.
- As the Dimensionality grows, fewer observations per region, ie density.
- Actual/True content is in the very small subset of this high dimensional space.
- More dense the space, better for the algorithm.

Dimensionality Reduction Algorithms

- How'd you identify highly significant variable(s) out 1000 or 2000?
- Feature selection: Select only relevant features (based on domain, correlation, etc)
- Feature Extraction: Compose new features based on combination of all existing ones.

Dimensionality Reduction Algorithms

- Principal Components Analysis (PCA) is a Feature Extraction method.
- Projects data from high-dimensional space into a lower-dimensional space

Dimensionality Reduction Example

Body Measurements

Say that we have a data-set of the following measurements from a large set of human volunteers with the following variables:

- height
- weight
- waist size
- shoe size
- length of right arm
- length of left arm

Body Measurements

- length of torso
- hat size
- left hand ring size
- right hand ring size

Technically we have 10 variables, though most of the variation in the data-set can be summarized by at most 2-3 variables.
What can you do? Domain knowledge?

Reduction in Dimensions

In decreasing order of variation, consider the following measurements that can be derived from these 10 variables

- height: captures a large amount of the variation in the total dataset.
- body mass index: should be relatively uncorrelated with overall height, captures much of the next largest variation in the data.
- ratio of torso length to total height: attempts to capture the remaining variation based on how height is distributed over a given individuals frame.

This is dimension reduction is based on domain knowledge, but is there anything generic? Yes, PCA (Principal Component Analysis).

PCA Algorithm

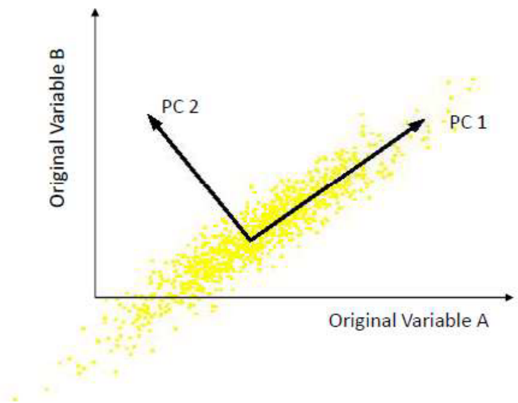
PCA

- One of the oldest (1901!)
- To understand "important" dimensions

PCA

- PCA looks at the data and tries to find a direction-line along which data is spread the most.
- Once done, you look for another (perpendicular) dimension which has second most variation.
- These are called as Principal components.
- So, if we TRANSFORM the data to the coordinate system formed by Principal Components, then as most of the variations are captured, a few axes capture most and thats the Dimension Reduction.

PCA in a nutshell



- Orthogonal directions of greatest variance in data
- Projections along PC1 discriminate the data most along any one axis

PCA

- First principal component is the direction of greatest variability (co-variance) in the data
- Second is the next orthogonal (uncorrelated) direction of greatest variability. So first remove all the variability along the first component, and then find the next direction of greatest variability
- And so on

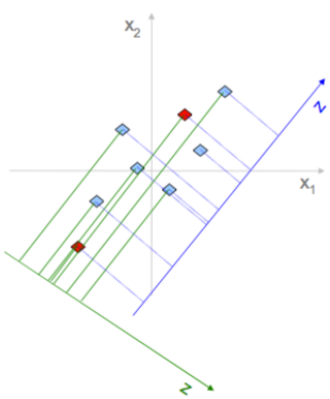
PCA

Principle

- Linear projection method to reduce the number of parameters
- Transfer a set of correlated variables into a new set of uncorrelated variables
- Map the data into a space of lower Dimensionality
- Form of unsupervised learning

Why greatest variability?

- Example: reduce 2-dimensional data to 1
- Data points in 2D x_1, x_2 space are represented by projection points on z axis, so, only 1-D distances.
- See how two red points on different z axes are ad different distances.
- The one with max variance preserves original intent-structure.
- So, blue Z is better!!



PCA in Steps

PCA in Steps

- Standardize the data.
- Compute Co-variance matrix.
- Obtain the Eigenvectors and Eigenvalues from the covariance matrix or correlation matrix, or perform Singular Value Decomposition.
- Sort eigenvalues in descending order and choose the k eigenvectors that correspond to the k largest eigenvalues where k is the number of dimensions of the new feature subspace ($k \leq d$).
- Construct the projection matrix W from the selected k eigenvectors.
- Transform the original dataset X via W to obtain a $k$dimensional feature subspace X'.

Standardizations

- Center the data by subtracting mean.
- Mathematically, this can be done by subtracting the mean and dividing by the standard deviation for each value of each variable.
  $z = \frac{value - mean}{standarddeviation}$
- Once the standardization is done, all the variables will be transformed to the same scale.
- Subtracting the mean makes variance and covariance calculation easier by simplifying their equations.
- The variance and co-variance values are not affected by the mean value.

Covariance Matrix Computation

- To see if there is any relationship between them.
- Variables are highly correlated in such a way that they contain redundant information.
- The covariance matrix is a $p \times p$ symmetric matrix (where p is the number of dimensions) that has as entries the covariances associated with all possible pairs of the initial variables
- For example, for a 3-dimensional data set with 3 variables x, y, and z, the covariance matrix is a 3×3 matrix of this from:

$$\begin{bmatrix} Cov(x,x) & Cov(x,y) & Cov(x,z) \\ Cov(y,x) & Cov(y,y) & Cov(y,z) \\ Cov(z,x) & Cov(z,y) & Cov(z,z) \end{bmatrix}$$

- Diagonal is same variable, and matrix is symmetric.

Whats Eigen Values/Vectors?

- Example: We are provided with 2-dimensional vectors $v_1, v_2, \ldots, v_n$.
- Then, if we apply a linear transformation T (a 2x2 matrix) to our vectors, we will obtain new vectors, called $b_1, b_2, \ldots, b_n$.
- $b_1, b_2, \ldots, b_n$ are obviously different in values and direction from $v_1, v_2, \ldots, v_n$ as we are transforming them (remember homogeneous transformation matrix?)
- But, there are special vectors, once applied the transformation T, they change length but not direction.
- Those vectors are called eigenvectors, and the scalar which represents the multiple of the eigenvector is called eigenvalue. $Tv_i = \lambda v_i$
- Thus, each eigenvector has a correspondent eigenvalue.
- if we consider our co-variance matrix C and collect all the corresponding eigenvectors into a matrix V (where the number of columns, which are the eigenvectors, will be equal to the number of rows of C), we will obtain something like that: $CV = VL$, where L is a diagonal matrix where all the eigenvalues (as many as the eigenvectors) are stored

$$\begin{bmatrix} Var(x) & Cov(x,y) \\ Cov(y,x) & Var(y) \end{bmatrix} \begin{bmatrix} \boldsymbol{v_1} \\ \boldsymbol{v_2} \end{bmatrix}' = \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} \begin{bmatrix} \boldsymbol{v_1} \\ \boldsymbol{v_2} \end{bmatrix}$$

Intuition
This is the mathematical heart of PCA: the eigenvectors of the covariance matrix are exactly the principal component directions, and their eigenvalues tell you how much variance each direction captures. Sorting eigenvectors by eigenvalue and keeping only the top few is precisely what "keeping the principal components" means.

(Ref: PCA: Eigenvectors and Eigenvalues - Valentina Alto)

Detour: How to Compute Eigen values and vectors

- Let A be an $n \times n$ matrix. A *non-zero* vector $\vec{x} \in \mathbb{R}^n$ is called an eigenvector of A if there exists some scalar $\lambda \in \mathbb{R}$ so that $A\vec{x} = \lambda\vec{x}$.
- If $\vec{x}$ is an eigenvector of A, the corresponding value λ is called an eigenvalue of A, and we say that λ is an eigenvalue of A with eigenvector $\vec{x}$.
- While an eigenvector $\vec{x}$ must be non-zero (so that we are always excluding the trivial case $A\vec{0} = \vec{0}$), it is possible for the value λ to be zero.
- If λ is an eigenvalue for A, the eigenvectors for A corresponding to λ along with $\vec{0}$ form a subspace of $\mathbb{R}^n$.

Definition 1.

$$\begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 6 \\ -5 \end{pmatrix} = \begin{pmatrix} -24 \\ 20 \end{pmatrix} = -4 \begin{pmatrix} 6 \\ -5 \end{pmatrix}.$$

Thus we see that $\begin{pmatrix} 6 \\ -5 \end{pmatrix}$ is an eigenvector of this matrix, and -4 is the corresponding eigenvalue.

Derivation

Definition 2. For any scalar λ,

$$\begin{aligned} A\vec{x} &= \lambda\vec{x} \\ &\Leftrightarrow A\vec{x} - \lambda\vec{x} = \vec{0} \\ &\Leftrightarrow A\vec{x} - \lambda I\vec{x} = \vec{0} \\ &\Leftrightarrow (A - \lambda I)\,\vec{x} = \vec{0} \\ &\Leftrightarrow \vec{x} \in (A - \lambda I) \end{aligned}$$

Derivation

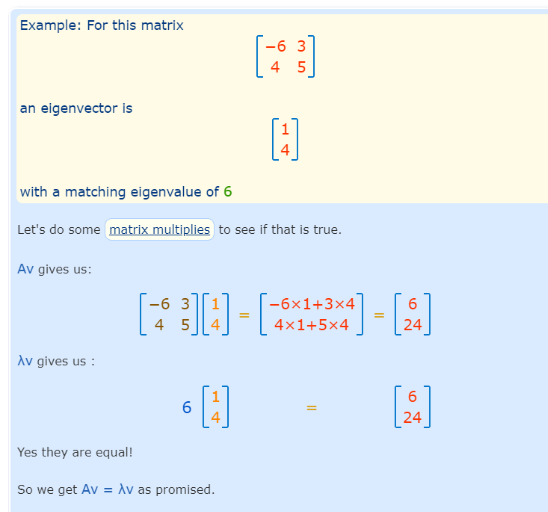
Theorem 3. *A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if λ satisfies the characteristic equation*

$$\det(A - \lambda I) = 0.$$

Theorem 4. $\det(A - \lambda I)$ *is a polynomial in λ of degree n. Hence, an $n \times n$ matrix has at most n eigenvalues.*

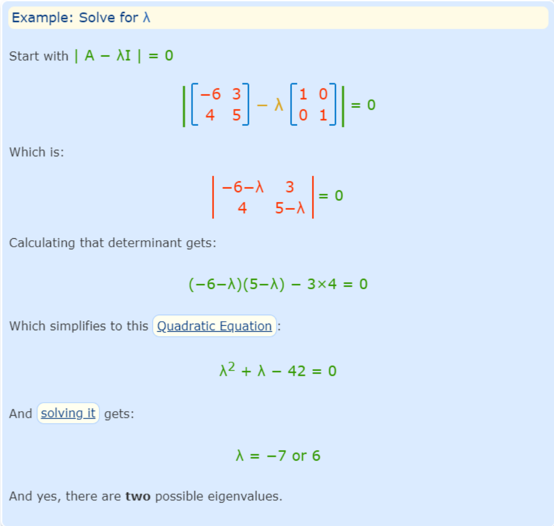
Intuition
Solving $\det(A - \lambda I) = 0$ is really asking: for which stretch factor λ does A have a direction it only scales, never rotates? Since a degree-n polynomial has at most n roots, an $n \times n$ matrix has at most n such special directions – exactly the eigenvectors PCA sorts by eigenvalue to find its principal components.

Example



(Ref: Eigenvector and Eigenvalue - Maths is fun)

Example



Example: Solve for λ

Start with $|A - \lambda I| = 0$

$$\left| \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

Which is:

$$\begin{vmatrix} -6-\lambda & 3 \\ 4 & 5-\lambda \end{vmatrix} = 0$$

Calculating that determinant gets:

$$(-6-\lambda)(5-\lambda) - 3\times 4 = 0$$

Which simplifies to this Quadratic Equation:

$$\lambda^2 + \lambda - 42 = 0$$

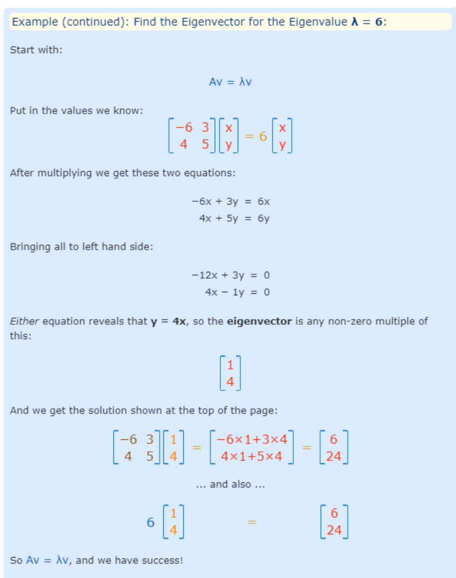
And solving it gets:

$$\lambda = -7 \text{ or } 6$$

And yes, there are **two** possible eigenvalues.

(Ref: Eigenvector and Eigenvalue - Maths is fun)

Example



Example (continued): Find the Eigenvector for the Eigenvalue **λ = 6**:

Start with:

$$Av = \lambda v$$

Put in the values we know:

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 6 \begin{bmatrix} x \\ y \end{bmatrix}$$

After multiplying we get these two equations:

$$\begin{aligned} -6x + 3y &= 6x \\ 4x + 5y &= 6y \end{aligned}$$

Bringing all to left hand side:

$$\begin{aligned} -12x + 3y &= 0 \\ 4x - 1y &= 0 \end{aligned}$$

Either equation reveals that **y = 4x**, so the **eigenvector** is any non-zero multiple of this:

$$\begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

And we get the solution shown at the top of the page:

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -6\times1+3\times4 \\ 4\times1+5\times4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

... and also ...

$$6 \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

So $Av = \lambda v$, and we have success!

Now it is your turn to find the eigenvector for the other eigenvalue of -7

(Ref: Eigenvector and Eigenvalue - Maths is fun)

Why?

What is the purpose of these?

- Wrt, transformation matrices, eigenvector is "the direction that doesn't change direction" !
- And the eigenvalue is the scale of the stretch:
  - 1 means no change,
  - 2 means doubling in length,
  - -1 means pointing backwards along the eigenvalue's direction
  - etc

(Ref: Eigenvector and Eigenvalue - Maths is fun)

Why 'Eigen'?

- Eigen is a German word meaning "own" or "typical": "das ist ihnen eigen" is German for "that is typical of them"
- Sometimes in English we use the word "characteristic", so an eigenvector can be called a "characteristic vector".

(Ref: Eigenvector and Eigenvalue - Maths is fun)

Back to PCA

Practical Scenario for PCA

If we want to reduce the dimensionality of our dataset? Imagine we start with 5 features and we want to handle 2 features instead. So, the procedure will be the following:

- Computing the C matrix our data, which will be 5x5
- Computing the matrix of Eigenvectors and the corresponding Eigenvalues
- Sorting our Eigenvectors in descending order
- Building the so-called projection or transformation matrix W, where the k eigenvectors we want to keep (in this case, 2 as the number of features we want to handle) will be stored.
- Hence, in our case, our W will be a 5x2 matrix (in general, it is a dxk matrix, where d =number of original features and k =number of desired features).
- Transforming the original data, which can be represented as an nxd matrix X (where n =number of observations and d =number of features), via the projection matrix W, obtaining a new dataset or matrix X' which will be nxk.

(Ref: PCA: Eigenvectors and Eigenvalues - Valentina Alto)

PCA Example Working

PCA Example in Steps

Subtract the mean

- From each of the data dimensions.
- All the x values have their $\bar{x}$ subtracted and y values have $\bar{y}$ subtracted from them.
- This produces a data set whose mean is zero.
- Subtracting the mean makes variance and covariance calculation easier by simplifying their equations.
- The variance and co-variance values are not affected by the mean value.

PCA Example in Steps

DATA:

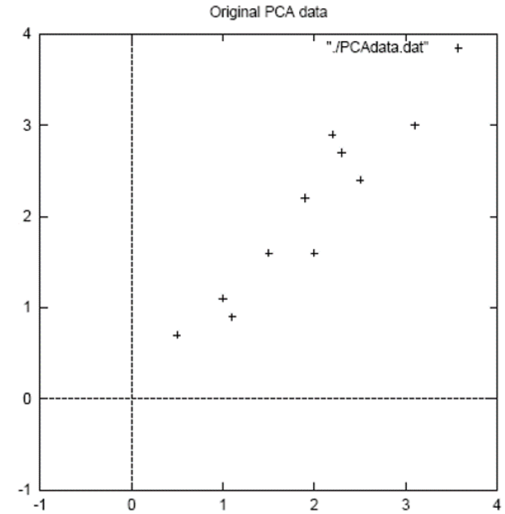
x	y
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3.0
2.3	2.7
2	1.6
1	1.1
1.5	1.6
1.1	0.9

ZERO MEAN DATA:

x	y
.69	.49
-1.31	-1.21
.39	.99
.09	.29
1.29	1.09
.49	.79
.19	-.31
-.81	-.81
-.31	-.31
-.71	-1.01

PCA Example in Steps

Original Data



PCA Example in Steps

- Calculate the co-variance matrix (Manually, Python, Matlab, etc)

cov = .616555556 .615444444
.615444444 .716555556

- Since the non-diagonal elements in this co-variance matrix are positive, we should expect that both the x and y variable increase together.

PCA Example in Steps

- Calculate the eigen-vectors and eigenvalues of the co-variance (Manually, Python, Matlab, etc)

eigenvalues = .0490833989, 1.28402771
eigenvectors = -.735178656 -.677873399,
.677873399 -.735178656

- Note they are perpendicular to each other

PCA Example in Steps

- Highest eigenvalue is the principle component.
- Here, it is pointed down the middle of the data.

- Order eigen vectors by eigenvalue, highest to lowest.
- The order of significance.
- Keep important ones.
- Small eigenvalues ignorable

PCA Example in Steps

DATA:

x	y
-.827970186	-.175115307
1.77758033	.142857227
-.992197494	.384374989
-.274210416	.130417207
-1.67580142	-.209498461
-.912949103	.175282444
.0991094375	-.349824698
1.14457216	.0464172582
.438046137	.0177646297
1.22382056	-.162675287

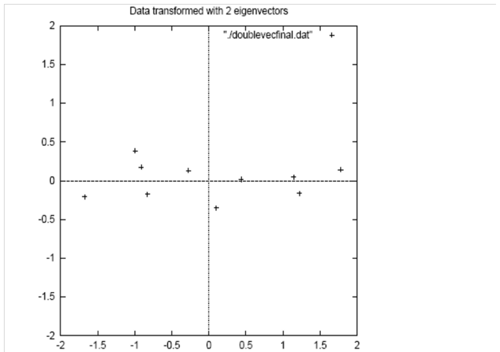
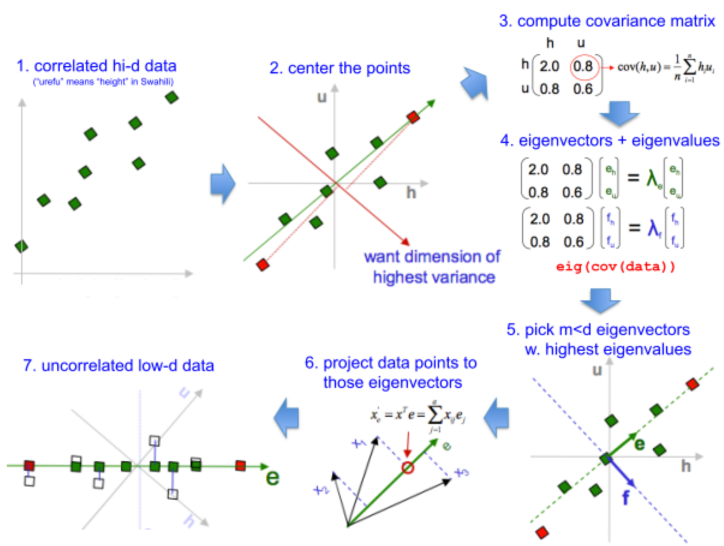


Figure 3.3: The table of data by applying the PCA analysis using both eigenvectors, and a plot of the new data points.

PCA in a nutshell

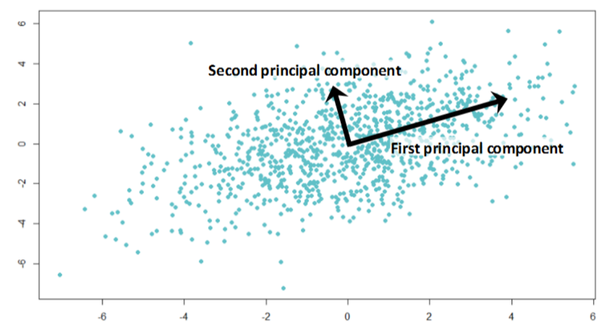


PCA

- A principal component is a normalized linear combination of the original predictors in a data set.
- The first principal component can be written as: $Z^1 = w^{11}X^1 + w^{21}X^2 + \ldots + w^{p1}X^p$
- It captures the maximum variance in the data set.
- It determines the direction of highest variability in the data.
- Larger the variability captured in first component, larger the information captured by component.
- No other component can have variability higher than first principal component.

PCA

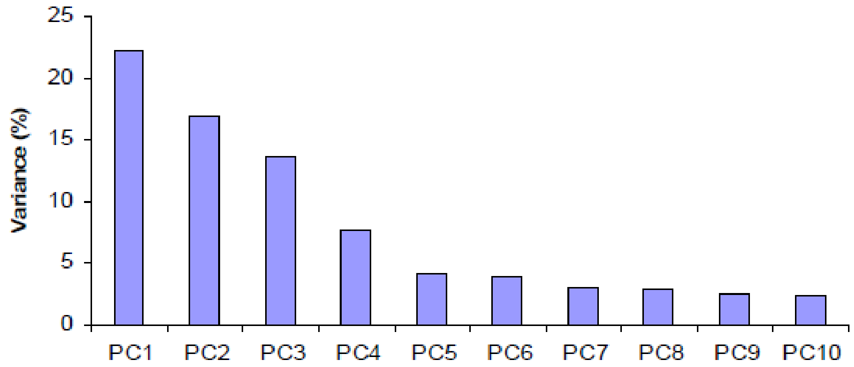
- The first principal component results in a line which is closest to the data i.e. it minimizes the sum of squared distance between a data point and the line.
- Similarly, we can compute the second principal component also. $Z^2 = w^{12}X^1 + w^{22}X^2 + \ldots + w^{p2}X^p$
- If the two components are uncorrelated, their directions should be orthogonal



- All succeeding principal component capture the remaining variation without being correlated with the previous component.

PCA

Can ignore the components of lesser significance.



You do lose some information, but if the eigenvalues are small, you don't lose much.

Principal Component Analysis

Advantages

- Useful Pre-processing step
- Reduces computational complexity
- Noise reduction

Principal Component Analysis

Limitations

- Linear manifold only
- Co-variance Matrix huge, finding eigen-vectors is slow
- SVD comes to rescue

A look ahead

- The main shortcoming of principal components are that they only capture global linear structures in the data. This tends to be a larger problem for prediction than it is for visualization.
- Figuring out how to get non-linear extensions of principal components is a wide open problem in statistic and machine learning. Some avenues of research include:
  - locally linear embedding
  - factor models
  - diffusion maps
  - mixture models